

INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi – 110016

INTERNSHIP REPORT

Arduino Based Embedded Systems

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Guide Name

Designation

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1 Introduction

This internship focused on the design and implementation of embedded systems using the Arduino UNO platform. The experiments cover basic digital interfacing, sensor integration, display modules, automation, and embedded applications.

The objective of the internship was to strengthen practical knowledge of microcontrollers, embedded programming, hardware interfacing, and real-time control systems through hands-on projects.

The report documents each experiment with its objective, circuit design, implementation methodology, working principle, results, and conclusion.

2 Arduino Fundamentals

2.1 LED Flash System

2.1.1 Objective

The objective of this experiment is to interface an LED with Arduino UNO and make it blink at a fixed interval.

2.1.2 Components Used

2.1.3 Circuit Diagram

2.1.4 Software Used

Arduino IDE

SimulIDE

2.1.5 Source Code

The complete Arduino source code is available online.

[Click here to view the source code](#)

2.1.6 Working Principle

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2.1.7 Conclusion

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